Macroeconomics I — MPE Insper

Mock Exam 5 (Simulado 5) — whole course — 2026, 3rd term

Instructions. Four questions, 10 points in total. Duration: 3 hours, closed book, non-programmable calculator allowed. Show every step; box final answers; sign every derivative before interpreting it; in diagrams label the axes, name every curve and mark the direction of every shift. Notation: Kurlat (2020) in Questions 1–3, Benigno (2015) in Question 4. All numbers are illustrative. Natural logarithms throughout.

Question 1. (2.0 points) Coffee, schools and saving in Minas Gerais.

Part A — measurement. One year in a coffee region, values in R$ million.

- A trading company imports fertiliser for 50 and sells all of it to a coffee farm for 80. It pays 10 in wages.
- The farm harvests coffee worth 300 at market prices. It sells 200 to a roaster, exports 60 to Germany, and the remaining 40 is still in its warehouse, unsold, at the end of the year. It pays 120 in wages.
- The roaster also imports packaging for 20 and sells all its roasted coffee, 450, to Brazilian households. It pays 150 in wages.
- The state government runs free public schools: it pays teachers 100 and buys nothing else.

Each firm's profit is whatever remains of its revenue after inputs and wages.

a) (1.0 point) (i) Compute the region's GDP by the product (value-added), income and expenditure approaches, and show that they agree. (ii) Next year the farm harvests nothing. It sells its 40 of stored coffee to the roaster, which roasts it and sells it to households for 90; nothing else happens in this chain. Compute the chain's contribution to next year's GDP by the product and the expenditure approaches. (iii) A columnist writes: "This year's GDP overstates the region's output, because 40 of coffee was never sold; and it understates it, because the schools charge nothing." Evaluate each half of the claim.

Solution:

Part B — saving and growth. The Ministry of Planning uses the Solow model in *per-worker* terms, with no technological progress, saving rate s, population growth n and depreciation δ constant:

$$y = Ak^{\alpha}, \tag{S1}$$

$$(1+n)k_{t+1} = (1-\delta)k_t + sAk_t^{\alpha}. \tag{S2}$$

Brazil is in steady state with $\alpha = 1/3$ and $s = 0.15$.

b) (1.0 point) (i) Derive steady-state capital k^* and consumption c^* per worker as functions of s, and find the saving rate s_{GR} that maximises c^*. Confirm it with a condition on the marginal product of capital. (ii) A pension reform raises the saving rate permanently from 0.15 to s_{GR} at date t_0. Compute the percentage change in consumption per worker on impact and between the two steady states, and the change in steady-state output per worker. (iii) Show graphically: the Solow diagram before and after the reform, and the time paths of y and c. (iv) Is the reform necessarily good for Brazilians? Who pays for it?

Solution:

Question 2. (2.5 points) A labour tax: what it does depends on what the government does with the money. A one-period economy has a fixed capital stock $\bar{K}$, owned by the representative household, which also owns the competitive firm. The household has one unit of time and utility (U), with $\psi = 1/3$. The firm produces (F) with $\alpha = 1/3$; normalise $A\bar{K}^{\alpha} = 1$. Labour income is taxed at rate τ, so the household's budget is (BC), where T is a lump-sum transfer that the household takes as given:

$$u = \ln c + \psi \ln(1 - L), \tag{U}$$

$$Y = A\bar{K}^{\alpha} L^{1-\alpha}, \tag{F}$$

$$c = (1 - \tau) wL + r\bar{K} + \Pi + T. \tag{BC}$$

Congress debates a 25% tax on labour income.

a) (0.7 point) Let $\tau = T = 0$. Write the household's and the firm's problems, their first-order conditions and the market-clearing conditions; show that profits are zero, and find the equilibrium L, Y, w and c. Then solve the problem of a planner who chooses c and L subject only to technology, and show that it gives the same allocation. State the theorem behind this and the assumptions it needs.

Solution:

b) (0.7 point) Now $\tau = 25\%$ and the government returns the whole revenue as the transfer, so $T = \tau w L$ in equilibrium. Find L (in general and in numbers). Explain why returning the money does not undo the effect of the tax. Show graphically, in the plane (leisure $\ell = 1 - L$, consumption c): the production possibilities frontier, the equilibrium of (a), the new equilibrium, and the household's budget line at the new equilibrium, with and without the transfer. Use the graph to explain why the new allocation is inefficient.

Solution:

c) (0.6 point) Instead, the same 25% tax finances government purchases $G = \tau w L$ that give households no utility ($T = 0$). Find L. Using (a) and (b), split the change in hours relative to $\tau = 0$ into a substitution effect and an income effect, and explain why hours fall less than in (b). What happens if $\alpha = 0$? An economist argues: "Labour taxes spent by the government barely change hours, so they cause almost no distortion." Evaluate, comparing with financing the same G through a lump-sum tax.

Solution:

d) (0.5 point) Return to the rebated tax of (b). Compute the welfare cost of the 25% tax as the percentage increase in consumption, at the allocation of (b), that would give households the utility of (a). Show that the cost is of second order in τ (that a small tax has a much smaller cost) and check it with $\tau = 12.5\%$. Which instrument would avoid the cost?

Solution:

Question 3. (2.5 points) The Real plan, the trip to the bank and the inflation tax. A stylised Brazil in the early 1990s. A representative household spends its real income Y per year at a constant rate. It keeps its wealth in an account that pays the nominal rate i and withdraws cash N times a year, in equal amounts; cash pays no interest. Each trip to the bank costs F in goods (queue, fee, time), that is pF in money. Take $Y = 60{,}000$ (reais of constant purchasing power per year) and $F = 25$. In the long run real output grows at $g = 4\%$ a year, the real interest rate is $r = 3\%$ and the money supply grows at $\mu = 11\%$ a year. Treat all money as currency (money multiplier equal to one) and use Kurlat's Fisher equation $i = r + \pi$.

a) (0.6 point) Derive the optimal number of trips N and the demand for real money balances M/p from the household's cost minimisation. Find the income and interest elasticities of money demand and the velocity of money. Compute N, M/p and velocity at $i = 12\%$. Show graphically how money demand depends on i, and how a fall in F (more ATMs) moves it.

Solution:

b) (0.7 point) Using money-market equilibrium $M = p\, m^D(Y, i)$, derive the steady-state inflation rate when Y grows at g and i is constant, and compute π and i. In mid-1994 the central bank cuts money growth permanently and credibly to $\mu = 5\%$. Find the new π and i and the change in real balances. Since the stock of money does not jump by itself, what must happen to the price level on the day of the reform? Which one-time change in M would avoid it? Draw $\ln M$ and $\ln p$ around the reform, with that one-time change. A commentator writes: "After the Real, M1 rose by tens of percent while inflation collapsed, so money does not cause inflation." Evaluate.

Solution:

c) (0.6 point) Seigniorage. Compute real seigniorage per household before and after the reform, and split it into the inflation tax and the part due to growth. Show that with the money demand of (a) seigniorage rises with μ for every μ, so there is no revenue-maximising money growth. Contrast with a Cagan semi-log demand $M/p = Ye^{-ai}$, $a > 0$: find the revenue-maximising μ and evaluate it for $a = 2$. Draw both curves. Which demand is consistent with Brazil's 1989–1993 experience, when money grew by more than 1,000% a year?

Solution:

d) (0.6 point) Costs of inflation. Compute the real resources spent on trips to the bank, in reais and as a share of Y, before and after the reform. Why is the forgone interest on cash not counted as a cost to society? Find the inflation and money-growth rates of the Friedman rule, and say what happens to the Baumol–Tobin solution there. Give two reasons why the central bank targets 3% rather than this rate.

Solution:

Question 4. (3.0 points) The Copom between a tax shock and a liquidity trap. Use Benigno (2015), in log deviations from an efficient steady state, with a short run and a long run:

$$y = \bar{y}_n + (g - \bar{g}) - \sigma\left[i - (\bar{p} - p) - \rho\right], \tag{AD}$$

$$p - p^e = \kappa(y - y_n), \tag{AS}$$

$$y_n = \frac{(1+\eta)a + \sigma^{-1}g - \mu}{\sigma^{-1} + \eta}, \tag{YN}$$

$$y_e = \frac{(1+\eta)a + \sigma^{-1}g}{\sigma^{-1} + \eta}. \tag{YE}$$

Initially $a = g = \bar{g} = \mu = 0$, $\bar{y}_n = 0$, $\bar{p} = p^e = 0$ and $i = \rho$. Take $\sigma = 1$, $\eta = 1$, $\kappa = 1/2$ and $\rho = 2\%$; y and p are in percent. The Copom sets i to minimise

$$L = \tfrac{1}{2}(y - y_e)^2 + \frac{\phi}{2\kappa}(p - p^e)^2, \qquad \phi > 0. \tag{LF}$$

a) (0.8 point) A temporary rise in a federal tax on firms' revenues raises the mark-up by $\mu = 4$ in the current period only. Find the changes in y_n and y_e and say which curve shifts. Derive the Copom's optimal targeting rule by substituting the relevant constraint into the loss, and the optimal y and p as functions of ϕ; evaluate them for $\phi = 4$. Explain why the Copom cannot reach $y = y_e$ and $p = p^e$ together here, while it could after a temporary productivity shock. Show graphically.

Solution:

b) (0.6 point) Find the policy rate i^* that implements the optimum, for general ϕ and for $\phi = 4$, and compare it with keeping $i = \rho$. Show that the Copom raises the rate if and only if $\phi > \sigma$. Interpret $\phi \to 0$, $\phi \to \infty$ and the knife edge $\phi = \sigma$, using the graph of (a).

Solution:

c) (0.7 point) Two years later the tax is gone ($\mu = 0$), but a household-debt crisis makes households expect lower long-run income: $\bar{y}_n$ falls by 4%, with y_n and y_e unchanged. Find the natural real rate r_n and the rate the Copom would like to set. Given $i \geq 0$, find y, p and the gaps when $i = 0$, and show the outcome graphically. What promise about $\bar{p}$ would take the economy to (y_e, p^e) with $i = 0$, and what does it require?

Solution:

d) (0.9 point) Instead of a promise, the government raises current spending g ($\bar{g}$ unchanged) while i stays at 0. Derive the multipliers dy/dg and $d(y - y_e)/dg$, evaluate them, find the g that closes the gap, and check what happens to p. Then suppose the Copom were away from the ZLB and followed its optimal rule: what would dy/dg be? Explain why the ZLB makes fiscal policy more powerful here. Does Benigno's §10 suggest even larger multipliers? Show the effect of g on the graph.

Solution:

End of exam.